



largest sensor size in a phone, which was only bettered 11 years later by Xiaomi Mi 11 Ultra with its 1/1.12-inch camera sensor.

While sensor sizes alone don't determine the quality of a camera phone, it remains a critical part of the experience. A larger sensor will be able to capture more light and pick up details that otherwise would be lost due to a smaller sensor. However, this has been a problem for most smartphone makers as large sensors often result in thicker phones. That's something neither the brands want nor the users.



The Nokia N8

In 2012, Nokia released the Nokia 808 PureView with a 41-megapixel camera that used an even larger 1/1.2-inch sensor with PureView Pro technology. It was our first look at pixel oversampling, or what is commonly known today as pixel binning. In this, the large image sensor combines information from multiple pixels into one single image pixel, which is sent to the onboard image processor that produces high-quality pictures with reduced noise. It also improves the low-light performance of the camera.



Xiaomi Mi 11 Ultra



Nokia 808 PureView

However, one of the major downsides of pixel binning is that the final processed image that we get is 1/4th the resolution of the sensor. To compensate ===for this, the image sensor had to be large enough for the camera to be able to produce a detailed picture with good exposure levels.

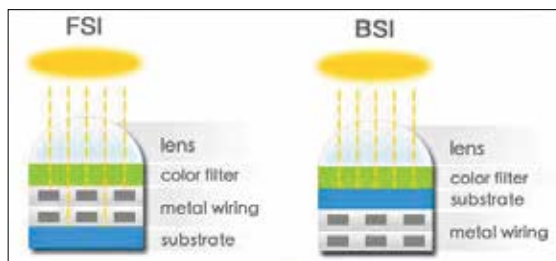
At the time, the smartphone industry was also going through a shift and established brands like Nokia were struggling to catch up to the Android and iOS smartphones. But this also gave rise to smartphone photography as more and more brands put their effort into camera development and found computational photography to be the next big area of focus.

For the next few years, smartphones continued to reign when it came to image quality. However, the overall progress in camera phone development slowed down, even though smartphones were getting more powerful every year. Later, when computational photography went mainstream, smartphones used multiple captures of a frame and an algorithm stitched it into a final image that had less noise. This technique greatly improved the image quality.

With the advent of BSI (Back-Side Illuminated) sensors, it became possible for smartphones to use larger sensors without increasing the thickness. This was a major breakthrough in smartphone photography in comparison to FSI (Front-Side Illuminated)

sensors where the sensor received limited light as it had to pass through filters and other layers, thereby losing signal information. On the other hand, BSI sensors allowed flexibility of industrial design, as they made room for lenses to be placed closer to the sensors, all without making the phone any thicker.

At the same time, we started seeing more than one camera in smartphones but for use cases that are way different from how we use multiple cameras today. Primarily, the first phones with dual cameras were made to capture 3D content because 3D was being perpetuated as the next big thing by consumer electronics brands. However, it failed to take off into something considerable and functional. From 2014, brands like HTC started playing with the idea of having a second camera as a depth sensor to capture depth information in a frame thereby allowing pictures to have a shallow depth of field effect. That is now known as the Portrait



How BSI sensors differ from FSI

mode in smartphone photography.

Soon, Apple jumped on the multiple camera bandwagon with the iPhone 7 Plus which had a 12-megapixel main camera along with a 12-megapixel telephoto camera. This enabled 2x optical zoom and brought portrait mode. Samsung followed suit with the Galaxy Note 8, its first flagship smartphone to feature dual cameras — a 12-megapixel telephoto lens with 3x optical zoom, similar to Apple's implementation.

In 2018, Huawei came up with the P20 Pro which is the first phone with three cameras co-developed by Leica, the iconic German imaging company. It had a 40-megapixel main camera with a 1/1.7-inch sensor, an 8-megapixel tel-